Reconstruction of Muon Tracks from BELLA Laser Plasma Accelerator

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Abstract

In this paper, we outline the development and use of tracking software and tools to track muons as part of the BELLA-μ experiment. The BELLA-μ project aims to use laser plasma accelerator (LPA) technology to create a muon source for muography imaging applications. To further this goal, we present here the use of six silicon particle detectors to track muons and characterize muon flux, energy, etc. Six ATLAS ITk pixel detector modules were set up in a telescope arrangement to detect and track muons produced by the BELLA (Berkeley Lab Laser Accelerator) LPA. We outline the development of tools to do tracking analysis and particle identification including software to read raw data from the detector modules and perform analysis tasks. Specifically, we use the Corryvreckan tracking software to load raw detector data, reconstruct particle tracks, and calculate muon momentum for the construction of a preliminary muon momentum spectrum.

I. INTRODUCTION

The muon is a fundamental particle classified as a lepton in the Standard Model. The muon is similar to the more familiar electron, which is also a lepton, although the muon mass is much greater. The muon mass is approximately $m_\mu = 105.66$ MeV which is roughly 207 times the mass of the electron [7]. This large mass allows muons to penetrate much further into matter and lose less energy to bremsstrahlung radiation. The penetration power of muons makes them ideal for nondestructive imaging of large and concealed objects. There are many muographic imaging applications including investigations of magma chambers of volcanoes and underground structures. The production of muon beams could also have applications in future muon-muon colliders or experiments like the Muon g-2 experiment at Fermilab [5].

However, all of these applications of muons are limited by the difficulty in producing muon beams. Physics research involving muon beams are currently done by colliding proton beams producing mesons which eventually decay into muons [5]. Imaging applications are even more difficult due to the immovability of large accelerators as well as the large imaging samples. Thus, the only available source of muons for imaging are cosmic ray muons. However, the cosmic ray muon flux is typically low and are thus limited in their applications.

To address the limitations of muon beam applications, the BErkeley Laboratory Laser Accelerator (BELLA) Facility at the Lawrence Berkeley National Laboratory is developing a muon source [5]. The BELLA facility makes use of laser plasma accelerator (LPA) technology to produces muons from the collision of an electron beam with a high-Z target. The BELLA LPA has demonstrated the efficacy of producing of high-flux GeV muons and experimental data has shown very good agreement with theory and simulations [5].

Further research into muon flux measurements and muon imaging is required for the BELLA-μ project. The BELLA-μ project is aims to create a compact, transportable source of GeV muons for a variety of imaging applications. For further study of the muon source, silicon tracking detectors have been selected. With these motivations, we present here the development of analysis tools for BELLA muon beams using an ATLAS ITk pixel detector telescope to track muons.

ATLAS is a large scale detector at the Large Hadron Collider in CERN dedicated to probing the Standard Model of particle physics, searching for new particle physics phenomena, and studying the newly discovered Higgs Boson. The ITk (Inner Tracker) is a new tracking detector designed for the ATLAS detector to prepare for the High Luminosity upgrade to the Large Hadron Collider (LHC) [4]. The ITk Pixel detector chip is the ITkPixV2 chip and is designed by the CERN RD53 collaboration to handle the increased performance requirements of the ATLAS ITk upgrade.

Using a six plane telescope configuration of these ITk pixel detector modules, muons can be identified by reconstructing tracks through the detectors. An analysis pipeline will be used with track reconstruction techniques to identify directional muons, measure their momentum, and analyze sample densities at the BELLA facility.

We specifically focus on the development process of tools for this analysis including the set up and implementation of track reconstruction software with the goal of achieving real time graphical track visualization. In addition to use for the BELLA-μ project, these analysis tools are especially useful as a test beam set-up for research and development for the ATLAS ITk pixel modules.

II. MATERIALS AND METHODS

A. Muon Production

In order to test the pixel detector telescope, we use BELLA muon beam data. Muons are produced at BELLA through pair production from high energy photons. A laser plasma accelerator is used to accelerate electrons to high energy which them emit the required high energy photons as *Bremsstrahlung* radiation upon interaction with heavy nuclei in the accelerator target. Bremsstrahlung radiation is emitted when high energy electrons are deviated from their path by the heavy "high-Z" atomic nuclei in the target. At the BELLA center, the high-Z target is composed of 16 inches of lead, 37 inches of steel, and 72 inches of concrete. The high energy Bremsstrahlung photons then interact with background atomic nuclei to pair produce charged leptons through the Bethe-Heitler process [5].

Leptons are a class of fundamental particles found on the Standard Model for Particle physics which includes both electrons and muons. If the energy of the Bremsstrahlung photons are higher than $2m_pc$ (where m_p is the mass of the lepton in question and c is the speed of light), pair production via this process can occur. Since the muon mass is much higher than the electron mass, electron beams in the accelerator (and thus the produced photons) must be at higher energy to produce muons. The resulting leptons are emitted along the Bremsstrahlung photon propagation direction in a weakly collimated beam.

In addition to Bethe-Heitler pair production, the accelerated electrons can interact with the heavy target to produce pions ($\pi^{\pm}$) which have a short lifetime. These pions may then decay into muons which contribute significantly to muon production at BELLA mostly at lower energy (0-3 GeV). The Bethe-Heitler process contributes more significantly at a higher energy range (1-10 GeV) [5].

B. ITk Pixel Detector and Data Readout

To detect the muons produced at the BELLA center, 6 ITk detector modules were used. Each detector module integrates a silicon sensor with the front-end readout chip (FE-chip) connected via bump bonding. A silicon sensor can essentially be thought of as a PN junction which is a combination of P and N doped substrates. Charged particles that pass through the sensor ionize matter along their path continuously losing energy. A bias voltage is applied to the PN junction so that the charges produced by the ionizing particle can be collected and sent to the readout chip as a signal via the bump bonds. An example image of an ITk readout chip is shown in figure 1.

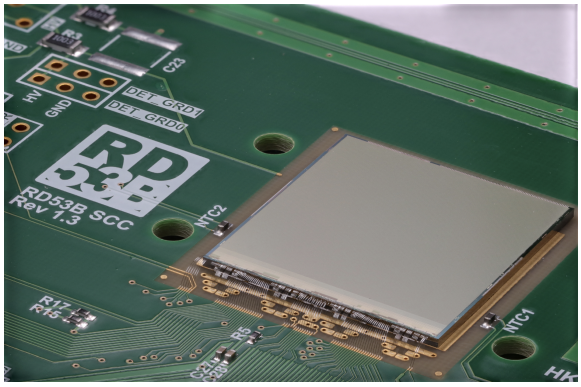


FIG. 1. Example of ATLAS Pixel Detector Chip on a readout card [3]

The ITkPixV2 readout chip consists of a 384 x 400 pixel matrix with each pixel being $50\mu m$ by $50\mu m$. The entire chip is $2cm$ by $2cm$ in size. When a particle passes through the detector module, "hits" may be read out in one or more pixels, which are identified by the pixel "row" and "column". The amount of charge collected in a pixel corresponds to voltage signal height which in turn corresponds to a "time over threshold" (ToT). This ToT is useful for determining the particle hit position by weighting pixels by ToT in the case that multiple pixels register hits. The row, column, and ToT information is read out by the ITkPixV2 chip and sent to a PCB (printed circuit board) to be sent to a computer.

C. Detector Module Telescope

The detector setup that we use at BELLA is called a telescope, where six of these ITk detector modules are arranged along the beamline as shown in figure 2. The six detector modules are arranged into two "stacks" of three detector module planes each (an "ingoing" stack and an "outgoing" stack). In between the two detector stacks, a $0.5T$ magnet is placed to conduct momentum measurements (discussed later). The entire telescope setup is surrounded by lead bricks to reduce background radiation which helps with muon identification.

D. YARR Data Acquisition

The detector setup makes use of the recently designed YARR (Yet Another Rapid Readout) data acquisition (DAQ) system. The YARR DAQ system is designed for flexible and efficient communication with readout chips via an FPGA board that enables high-speed data transmission [2]. The system enables most data processing to move from the FPGA to computer software in everyday com-

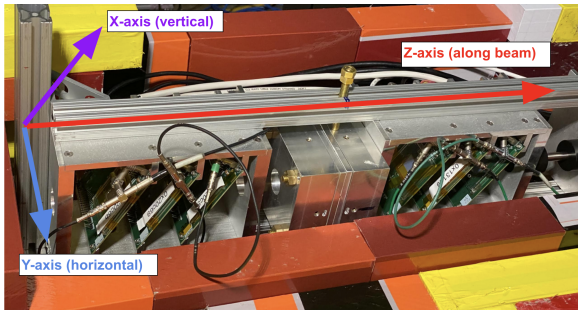


FIG. 2. Image of the BELLA beam telescope with 6 ITk detector modules. A $0.5T$ magnet is placed in the middle of the telescope for momentum measurement. The coordinate system used in our tracking framework is overlayed with the Z axis representing the beam axis. The X and Y axes represent the vertical and horizontal axes respectively (note unconventional axes used in reconstruction software).

puters which increases flexibility and portability by cutting the reliance on hardware for readout speed. The YARR DAQ system conveniently outputs data read from our six ITk detector modules as a raw binary file which can then be loaded into our tracking software for track reconstruction and analysis.

III. RESULTS AND DISCUSSION

A. The Corryvreckan Framework

The primary goal of this project was to setup a tracking and data analysis pipeline for BELLA muons using the Corryvreckan track reconstruction and analysis software [1]. Corryvreckan is a highly configurable software following a modular concept originally intended for offline accelerator test beam analysis. The software makes use of so called "modules" to perform various tasks including event building (from raw data), hit clustering, track reconstruction, and track analysis. Corryvreckan can be easily run from the command line by providing user generated configuration files specifying detector configurations, detector specifications, and analysis module parameters. The first step in a typical Corryvreckan analysis pipeline is to load in raw data and arrange them into "events" with associated start and end times as well as hit information. Using a Corryvreckan clustering module, groups of hits can then be organized into "clusters" which represent a single particle passing through a detector module. These clusters can further be organized into tracks passing through multiple detector module planes using a Corryvreckan track reconstruction module. An example diagram of a track is shown in figure 3. Then, further analysis modules are run making use of these hit, cluster, and track objects.

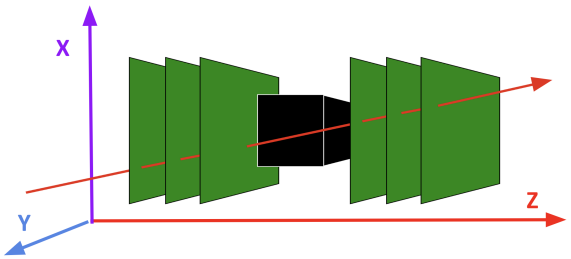


FIG. 3. 3D diagram of the BELLA detector module telescope with an example track passing through all six detector module planes. The coordinate axis used by the Corryvreckan software is shown with the z axis representing the beam axis.

As one of our primary track reconstruction tools, it is crucial to allow for compatibility between Corrvreckan and the raw data files generated by the YARR DAQ system. Thus, a large part of this project involved the development of the "EventLoaderYarr" Corryvreckan module in C++. This module reads in raw binary data in the YARR format provided a path to the directory containing the datafiles and associates each with the correct detector module plane. The module then creates events divided into $25ns$ *snapshots* or *triggers*. Each trigger may then contain particle hits which are read and turned into Corryvreckan hit objects containing the row, column, and ToT information previously described in section II B.

The development of this module for event loading into Corryvreckan now allows for seamless integration with YARR data acquisition. The preexisting Corrvreckan modules for hit clustering, track reconstruction, and track analysis can then be used on the data.

B. Detector Alignment

Once we able able to successfully identify tracks, we are able to obtain alignment information about the detector planes. To accurately reconstruct tracks in 3D space, the detector module planes must be perfectly aligned along the z-axis (beam axis). Even a small misalignment of even a fraction of a millimeter may correspond to a shift of multiple pixels. An exaggerated view of detector misalignment is shown in figure 4. To address this misalignment, the Corryvreckan software utilizes a reconstruction of the detector plane positions in a 3D space. The detector position and Euler angles fully specify the detector plane geometry. These positions and Euler angles can be adjusted in the tracking software before track reconstruction to minimize misalignment. This is done with preexisting alignment modules in the Corryvreckan software.

The alignment procedure in Corryvreckan is

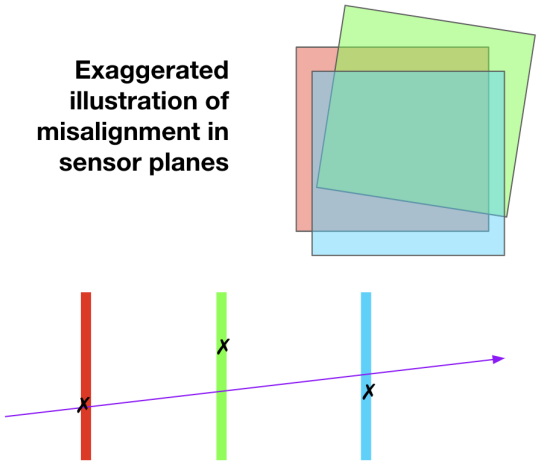


FIG. 4. An exaggerated illustration of misalignment between 3 detector planes. The top illustration shows the view of the detector planes face on as corresponding to the point of view of an incoming particle. The bottom shows a side view and includes x marks indicating where hits occur on one axis. An example of a reconstructed track is shown as the purple arrow. The hits don't match perfectly with the reconstructed track as a result of the misalignment.

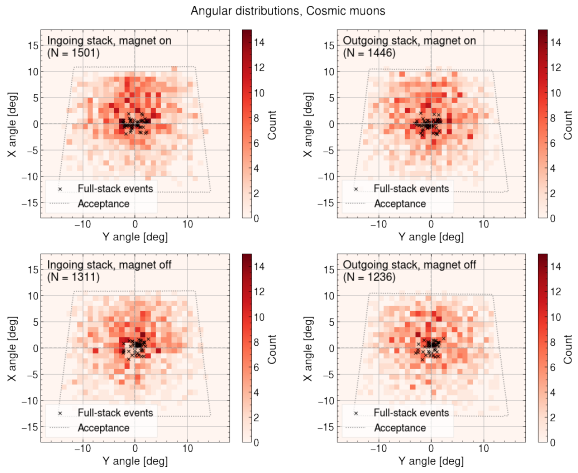


FIG. 5. Track angle distributions of cosmic ray muons. Our coordinate system is a three dimensional cartesian one so track angles are specified relative to the vertical axis (x angle) and horizontal axis (y angle). The left two plots correspond to tracks in the incoming detector stack and the right two correspond to the outgoing stack. The top two plots correspond to data taken with the *magnet in* while the bottom correspond to the *magnet out*. The small black x marks correspond the angles of full stack events. An outline of the calculated telescope acceptance is overlayed as a dotted line.

an iterative procedure which repeatedly identifies *straight-line* tracks and adjusts detector positions and Euler angles to minimize track χ^2/ndf and track residuals. Track residuals represent the differences between actual hits on the detector plane (the x marks in figure 4) and the reconstructed track intersection point with the detector plane. From these residuals, a χ^2/ndf value can be calculated representing the quality of the track reconstruction. Ideally, a distribution of track residuals (for all identified tracks) should follow a gaussian distribution centered on zero with a width corresponding to the spatial resolution of the pixels. Examples of residual distributions are shown in figure 6.

The alignment procedure also naturally requires the use of straight line tracks as tracks that bend would lead to uncertainty in the residuals. For this reason, data for alignment is taken without the magnet in the middle of the two detector stacks (see section II C). The magnet is required for momentum measurement but it does this measurement by bending the tracks of the charged muons. Thus, during all data taking campaigns, a portion of the data taking includes data with no magnet to provide alignment information.

C. Angular Acceptance

Furthermore, it was discovered that alignment is difficult because of statistical limitations. Identified tracks can be divided into "single stack events" and "full stack events", which correspond to a particle passing through only one of the two detector stacks and a particle passing through both detector stacks (all six detector planes) respectively. As a result of the spatial acceptance of the two detector stacks, it is much more likely for a muon to pass through only one stack. Passing through both detector stacks (which are separated by a distance of about $23cm$) would require the incoming muons to have very small angular deviation from the telescope z-axis. Plots of the 2D track angle distributions for cosmic muon data are shown in figure 5. The distributions of identified track angles (with respect to the telescope z-axis) shows that for a muon to pass through a single detector stack, the angle must withing around ± 12 degrees. The full stack tracks (black x marks on plot) have a much smaller angular distribution and have an acceptance of about ± 4 degrees. As a result of this smaller acceptance for full stack events, we have better statistics in each single stack individually.

D. Statistical Considerations

Data taking was split into two campaigns. One where the detector module telescope was placed horizontally aligned with the BELLA beamline to detect muons produced by the BELLA muon source. Another campaign involved the measurement of cosmic ray muons as calibration data and had the telescope placed vertically pointed at the

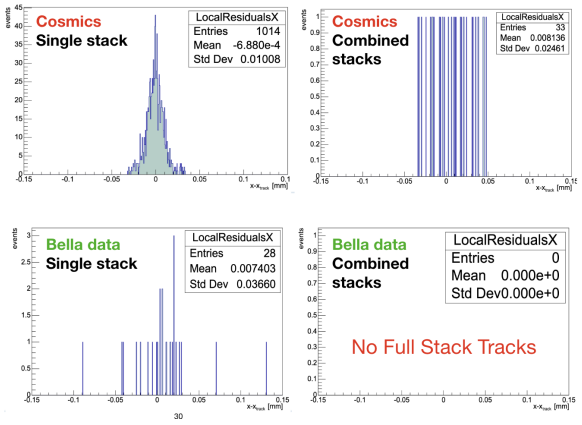


FIG. 6. Residual distributions for the two data taking campaigns (Cosmic muons on top and BELLA muon source data on the bottom). The left two plots show residual distributions for single stack events and the right two show full stack events. There were no full stack events in the BELLA muon source alignment data.

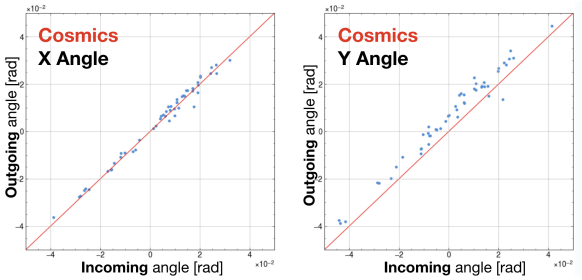


FIG. 7. Plots showing the incoming track angle vs the outgoing track angle for full stack events in the cosmic alignment data. Since in a perfectly aligned telescope, the incoming and outgoing track angle would be equal, the data points should be distributed along the y=x line shown in red. The left and right plots show track X angles (vertical telescope axis) and track Y angles (horizontal telescope axis) respectively. The deviation from the y=x line in the Y angle distribution on the right indicate a misalignment of the ingoing and outgoing detector stacks.

sky. The cosmic muon data was taken over a period of about 49 days while the BELLA beam muon source data was over a much shorter period of time. The BELLA muon source data spanned roughly 350 electron beam shots over a few days. Although the BELLA beam data was found to produce muons at a significantly higher rate than cosmic data, statistics were sparser than the cosmic data due to the large campaign period of the cosmic muon data taking.

As a result of the low statistics while taking BELLA muon source data, not enough tracks were identified when taking alignment data with the magnet out. This meant that the alignment procedure needed to be carried out on cosmic muon data which provided significantly more tracks (both single stack and full stack). Examples of the residual distributions used for the alignment procedure are shown in figure 6. As seen in the residual plots, the highest number of tracks were found in the cosmic muon data of which single stack tracks constituted the majority of tracks, making it an ideal starting point for the Corryvreckan alignment procedure. The use of single stack tracks means that even after aligning each detector module stack individually, there may still be misalignment between the two stacks themselves. However, we will soon see that this can be corrected for in the momentum calculation separately.

E. Momentum Calculation

After properly aligning the detector planes in the telescope, accurate track information including their angles with respect to the telescope z-axis can be exported from Corryvreckan. Tracks are identified separately in each detector stack so that full stack events will have a separate angle in the incoming and outgoing stacks. For momentum measurement, we take advantage of the fact that muons are charged particles whose trajectories will bend in the presence of a magnetic field. Thus we expect muon track angles for full stack events (in data *with magnet in*) to be different in the incoming stack and outgoing stack. The difference in this track angle between the two detector stacks corresponds to a bending angle α.

$$\alpha = \theta_{ingoingstack} - \theta_{outgoingstack} - c_{offset} \quad (1)$$

In addition to the difference in the two angles, a correction factor c_{offset} must be taken into account to correct for an angular misalignment between the two stacks. Since the detector planes in the two stacks were aligned separately, the stacks themselves may have some misalignment. This misalignment can be seen in the average track angles of cosmic muon alignment data as shown in figure 7. This misalignment is on the order of about 6mrad and is easy to correct for.

From the bending angle α, the muon momentum can be calculated using the following equation [6].

$$Br = 3.3356pc \quad (2)$$

Here, B represents the strength of the magnetic field, r represents the radius of curvature, p represents the muon momentum, and c the speed of light. With some minor approximations, the momentum can be calculated in GeV using this method. The radius of curvature can be approximated using the length of the magnetic field that the charged particle passes through (0.05m in our case) and the bending angle α as follows.

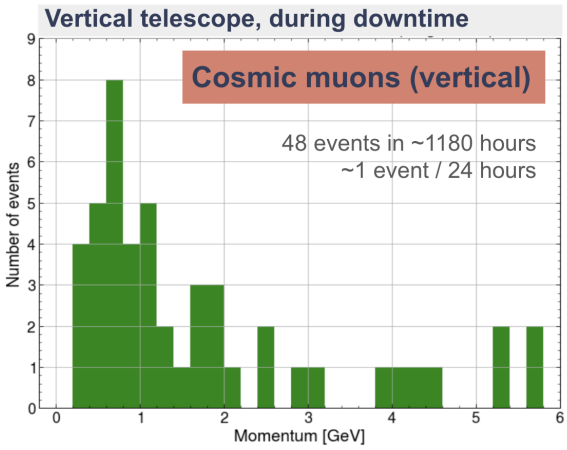


FIG. 8. Mometum spectra of cosmic ray muons identified when the detector telescope was configured vertically pointed at the sky. Momenta lower than around 0.2 GeV cannot be resolved as low momentum muons would bend too much to be detected by both detector stacks. Momenta higher than about 6 GeV also cannot be resolved since muons of high momentum will have very small bending angle whose measurement is limited by alignment.

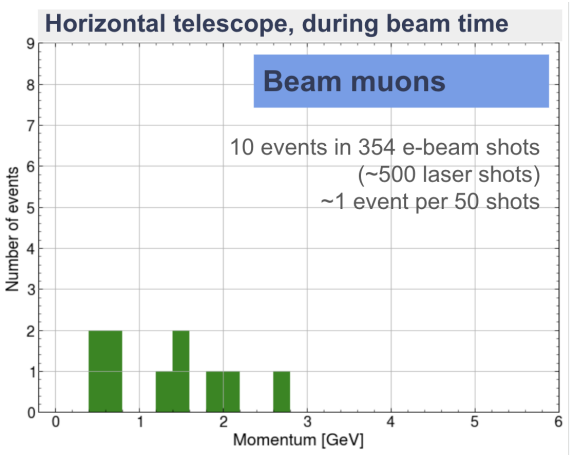


FIG. 9. The first momentum spectra of ten muon tracks identified from the BELLA muon source. Telescope was configured horizontally facing the BELLA beam line.

$$r \approx \frac{0.05}{sin(\alpha)} \tag{3}$$

From these simple calculations, we are able to calculate muon momentum for full stack muon tracks for both cosmic muons and BELLA beam muons. The track reconstruction work of this project thus culminated in the creation of a momentum spectrum using the cosmic muon calibration data and the first momentum measurements of muons produced by the BELLA muon source. These momentum spectra are are shown in figures 8 and 9 respectively. Although availability of data points are low for both spectra, these measurements show preliminary agreement with simulations.

IV. CONCLUSION

Through this project, we have been able to develop tools for muon tracking for the BELLA-μ project. The tools include a tracking analysis pipeline starting with track reconstruction in the Corryvreckan software and ending with python tools to calculate momentum from muon tracks. In doing so, we have been able to provide the first momentum calculations for muons produced by the BELLA muon source, which serves as an important proof of concept. One of the primary limitations of our analysis work was of low statistics. A large number of muon tracks are required since detector plane alignment and momentum resolution are both highly dependent on statistics. To further this project in the future, improving detector acceptance by using larger ITk detector modules is an important step. In addition to addressing the needs of the BELLA-μ project, the development of the Corryvreckan EventLoaderYarr module will be helpful for test beam measurements for testing the ITk pixel detectors.

V. ACKNOWLEDGMENTS

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